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JEE MAIN · QUESTION PAPER

JEE Main 2021 (Online) 24th February Evening Shift

JEE Main Question Paper with Answer Key

24 Feb 2021	90	22	English
HELD ON	QUESTIONS	OUT OF SYLLABUS	LANGUAGE

INSIDE THIS PDF

Chemistry	22 questions
Mathematics	19 questions
Physics	27 questions
Out of syllabus	22 questions
Answer key	at the end

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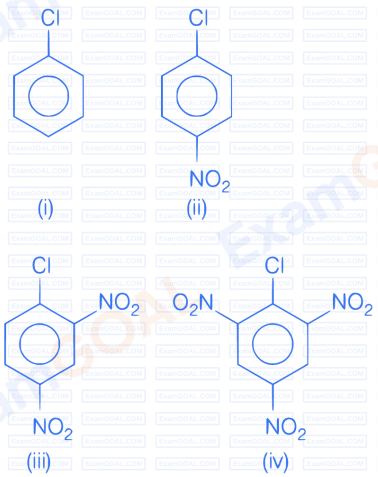
1. **Solve** — Print it, or work through it on screen.
2. **Check** — The answer key is on the last pages.
3. **Understand** — Scan the QR code beside any question for its full solution.

Chemistry 22 questions

Q1. MCQ (Single Correct Answer)

Open solution

The correct order of the following compounds showing increasing tendency towards nucleophilic substitution reaction is :



(A) (iv) < (i) < (ii) < (iii)

(B) (iv) < (iii) < (ii) < (i)

(C) (iv) < (i) < (iii) < (ii)

(D) (i) < (ii) < (iii) < (iv)

Q2. MCQ (Single Correct Answer)

Open solution

The **incorrect** statement among the following is :

(A) RuO_4 is an oxidizing agent

(B) Cr_2O_3 is an amphoteric oxide.

(C) $VOSO_4$ is a reducing agent

(D) Red colour of ruby is due to the presence of Co^{3+}

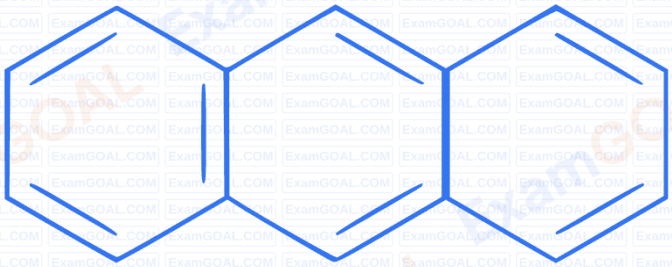
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Q3. MCQ (Single Correct Answer)

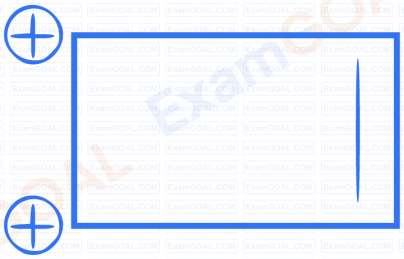
Open solution

Which one of the following compounds is non-aromatic?

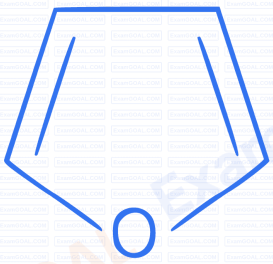
(A)



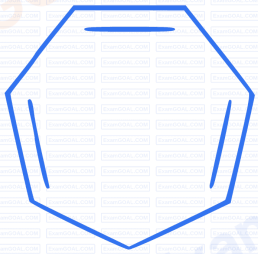
(B)



(C)



(D)



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Q4. MCQ (Single Correct Answer)

Open solution

According to Bohr's atomic theory :

(A) Kinetic energy of electron is $\propto \frac{Z^2}{n^2}$.

(B) The product of velocity (v) of electron and principal quantum number (n), $'vn' \propto Z^2$.

(C) Frequency of revolution of electron in an orbit is $\propto \frac{Z^3}{n^3}$.

(D) Coulombic force of attraction on the electron is $\propto \frac{Z^3}{n^4}$.

Choose the most appropriate answer from the options given below :

(A) (A), (C) and (D) only

(B) (A) only

(C) (C) only

(D) (A) and (D) only

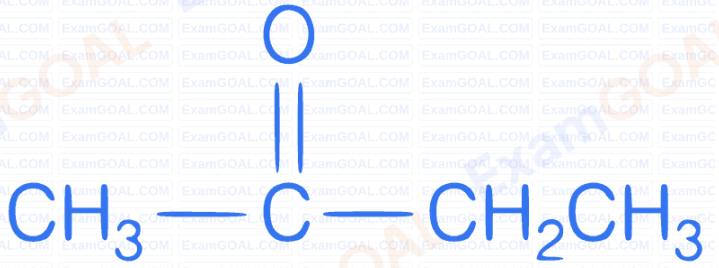
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Q5. MCQ (Single Correct Answer)

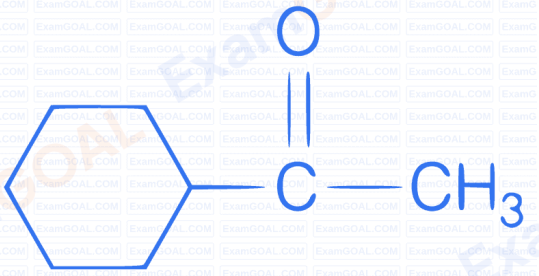
Which one of the following carbonyl compounds cannot be prepared by addition of water on an alkyne in the presence of $HgSO_4$ and H_2SO_4?

Open solution

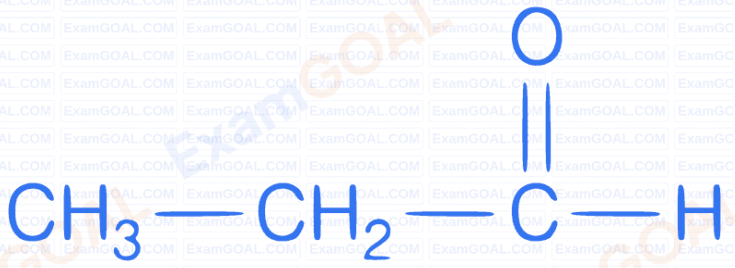
(A)



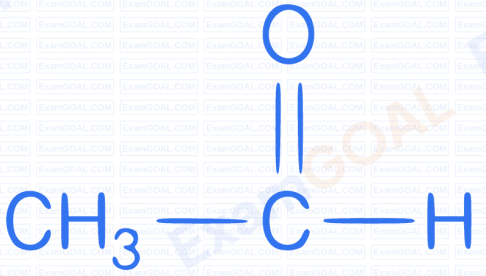
(B)



(C)



(D)



Q6. MCQ (Single Correct Answer)

The correct set from the following in which both pairs are in correct order of melting point is :

Open solution

(A) LiCl > LiF ; NaCl > MgO

(B) LiCl > LiF ; MgO > NaCl

(C) LiF > LiCl ; NaCl > MgO

(D) LiF > LiCl ; MgO > NaCl

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Q7. MCQ (Single Correct Answer)

What is the correct sequence of reagents used for converting nitrobenzene into m-dibromobenzene?

Open solution



(A) $\xrightarrow{NaNO_2}$ / $\xrightarrow{HCl}$ / $\xrightarrow{KBr}$ / $\xrightarrow{H^+}$

(B) $\xrightarrow{Sn/HCl}$ / $\xrightarrow{KBr}$ / $\xrightarrow{Br_2}$ / $\xrightarrow{H^+}$

(C) $\xrightarrow{Br_2/Fe}$ / $\xrightarrow{Sn/HCl}$ / $\xrightarrow{NaNO_2/HCl}$ / $\xrightarrow{CuBr/HBr}$

(D) $\xrightarrow{Sn/HCl}$ / $\xrightarrow{Br_2}$ / $\xrightarrow{NaNO_2}$ / $\xrightarrow{NaBr}$

Q8. MCQ (Single Correct Answer)

The calculated magnetic moments (spin only value) for species $[FeCl_4]^{2-}$, $[Co(C_2O_4)_3]^{3-}$ and MnO_4^{2-} respectively are :

Open solution

(A) 5.92, 4.90 and 0 BM

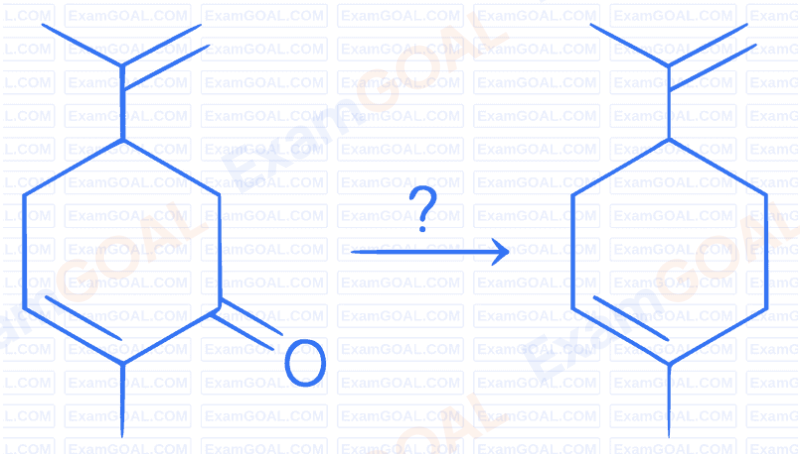
(B) 4.90, 0 and 1.73 BM

(C) 5.82, 0 and 0 BM

(D) 4.90, 0 and 2.83 BM

Q9. MCQ (Single Correct Answer)

Open solution



Which of the following reagent is suitable for the preparation of the product in the above reaction?

(A) Red P + Cl_2

(B) $NH_2 - NH_2/C_2H_5\overset{\ominus}{O}\overset{\oplus}{N}a$

(C) $NaBH_4$

(D) Ni/H_2

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Q10. MCQ (Single Correct Answer)

What is the correct order of the following elements with respect to their density?

(A) Cr < Zn < Co < Cu < Fe

(B) Cr < Fe < Co < Cu < Zn

(C) Zn < Cr < Fe < Co < Cu

(D) Zn < Cu < Co < Fe < Cr

Open solution

Q11. MCQ (Single Correct Answer)

Match List - I and List - II.

	List I		List II
(a)	Valium	(i)	Antifertility drug
(b)	Morphine	(ii)	Pernicious anaemia
(c)	Norethindrone	(iii)	Analgesic
(d)	Vitamin B_{12}	(iv)	Tranquilizer

(A) (a)-(i), (b)-(iii), (c)-(iv), (d)-(ii)

(B) (a)-(iv), (b)-(iii), (c)-(ii), (d)-(i)

(C) (a)-(ii), (b)-(iv), (c)-(iii), (d)-(i)

(D) (a)-(iv), (b)-(iii), (c)-(i), (d)-(ii)

Open solution

Q12. MCQ (Single Correct Answer)

The correct shape and $I-I-I$ bond angles respectively in I_3^- ion are :

(A) Linear; 180°

(B) T-shaped; 180° and 90°

(C) Trigonal planar; 120°

(D) Distorted trigonal planar; 135° and 90°

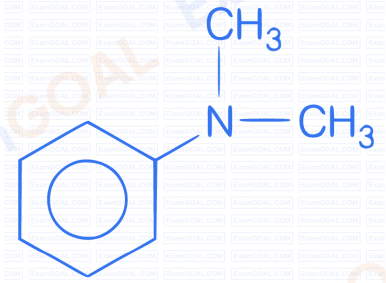
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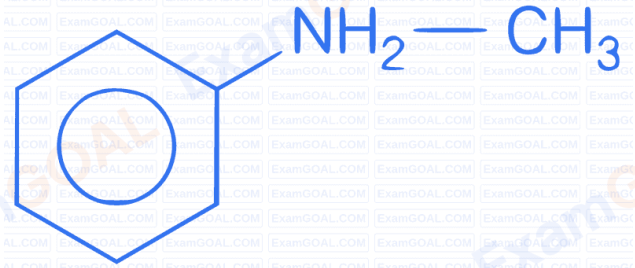
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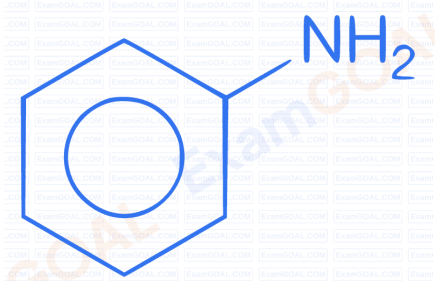
Q13. MCQ (Single Correct Answer)

Open solution

The diazonium salt of which of the following compounds will form a coloured dye on reaction with β-Naphthol in NaOH?

(A) 

(B) 

(C) 

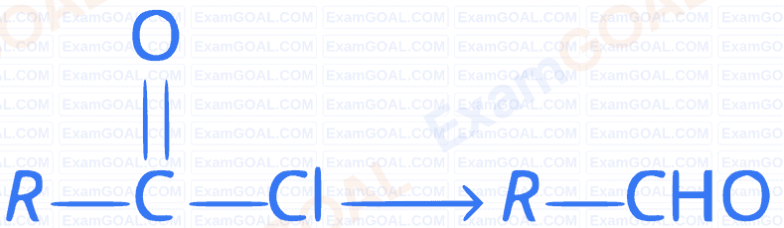
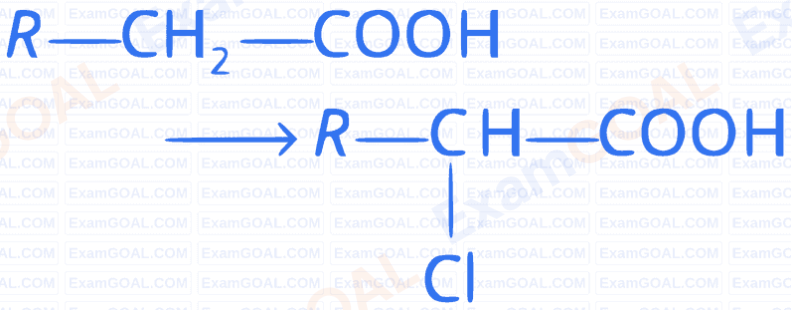
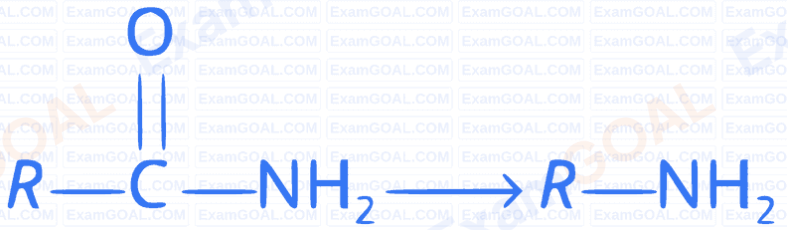
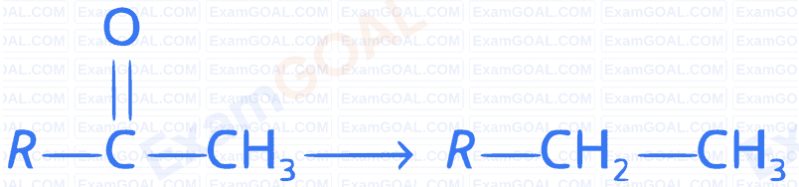
(D) 

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Q14. MCQ (Single Correct Answer)

Match List - I and List - II.

Open solution

	List-I		List-II
(a)		(i)	$Br_2/NaOH$
(b)		(ii)	$H_2/Pd - BaSO_4$
(c)		(iii)	$Zn(Hg)/H$ Conc. HCl
(d)		(iv)	Cl_2 Red P, H_2O

Choose the correct answer from the options given below :

(A) (a)-(ii), (b)-(iv), (c)-(i), (d)-(iii)

(B) (a)-(iii), (b)-(iv), (c)-(i), (d)-(ii)

(C) (a)-(iii), (b)-(i), (c)-(iv), (d)-(ii)

(D) (a)-(ii), (b)-(i), (c)-(iv), (d)-(iii)

Q15. Numerical

The formula of a gaseous hydrocarbon which requires 6 times of its own volume of O_2 for complete oxidation and produces 4 times its own volume of CO_2 is C_xH_y. The value of y is _____.

Open solution

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Q16. Numerical

The solubility product of PbI_2 is 8.0×10^{-9}. The solubility of lead iodide in 0.1 molar solution of lead nitrate is $x \times 10^{-6}$. mol/L. The value of x is _____. (Rounded off to the nearest integer)
[Given $\sqrt{2}$ = 1.41]

Open solution

Q17. Numerical

1.86 g of aniline completely reacts to form acetanilide. 10% of the product is lost during purification. Amount of acetanilide obtained after purification (in g) is _____ $\times 10^{-2}$.

Open solution

Q18. Numerical

Assuming ideal behaviour, the magnitude of log K for the following reaction at 25°C is $x \times 10^{-1}$. The value of x is _____. (Integer answer)

$$3HC \equiv CH_{(g)} \rightleftharpoons C_6H_{6(l)}$$

[Given : $\Delta_f G^o(HC \equiv CH) = -2.04 \times 10^5$ J mol^{-1} ; $\Delta_f G^o(C_6H_6) = -1.24 \times 10^5$ J mol^{-1} ; R = 8.314 J K^{-1} mol^{-1}]

Open solution

Q19. Numerical

C_6H_6 freezes at 5.5°C. The temperature at which a solution of 10g of C_4H_{10} in 200g of C_6H_6 freeze is _____ °C. (The molal freezing point depression constant of C_6H_6 is 5.12°C/m.)

Open solution

Q20. Numerical

Sucrose hydrolyses in acid solution into glucose and fructose following first order rate law with a half-life of 3.33 h at 25°C. After 9 h, the fraction of sucrose remaining is f. The value of $\log_{10}\left(\frac{1}{f}\right)$ is _____ $\times 10^{-2}$. (Rounded off to the nearest integer)

[Assume : ln 10 = 2.303, ln 2 = 0.693]

Open solution

Q21. Numerical

The magnitude of the change in oxidising power of the MnO_4^-/Mn^{2+} couple is $x \times 10^{-4}$ V, if the H^+ concentration is decreased from 1M to 10^{-4} M at 25°C. (Assume concentration of MnO_4^- and Mn^{2+} to be same on change in H^+ concentration). The value of x is _____.
$\left[Given : \frac{2.303RT}{F} = 0.059\right]$

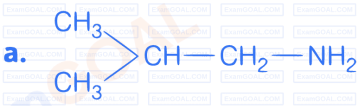
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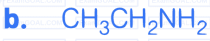
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Q22. Numerical

The total number of amines among the following which can be synthesized by Gabriel synthesis is _____.

Open solution

a. 

b. 

c. 

d. 

Mathematics 19 questions

Q23. MCQ (Single Correct Answer)

Let A and B be 3 $\times$ 3 real matrices such that A is symmetric matrix and B is skew-symmetric matrix. Then the system of linear equations $(A^2B^2 - B^2A^2)$ X = O, where X is a 3 $\times$ 1 column matrix of unknown variables and O is a 3 $\times$ 1 null matrix, has :

Open solution

(A) no solution

(B) exactly two solutions

(C) infinitely many solutions

(D) a unique solution

Q24. MCQ (Single Correct Answer)

The value of the integral, $\int_1^3 [x^2 - 2x - 2]dx$, where [x] denotes the greatest integer less than or equal to x, is :

Open solution

(A) -5

(B) $-\sqrt{2} - \sqrt{3} + 1$

(C) -4

(D) $-\sqrt{2} - \sqrt{3} - 1$

Q25. MCQ (Single Correct Answer)

Let a, b$\in$R. If the mirror image of the point P(a, 6, 9) with respect to the line

$\frac{x-3}{7} = \frac{y-2}{5} = \frac{z-1}{-9}$ is (20, b, $-$a$-$9), then | a + b |, is equal to :

Open solution

(A) 88

(B) 90

(C) 86

(D) 84

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Q26. MCQ (Single Correct Answer)

Open solution

For the system of linear equations:

$x - 2y = 1, x - y + kz = -2, ky + 4z = 6, k \in R,$

consider the following statements :

(A) The system has unique solution if $k \neq 2, k \neq -2$.

(B) The system has unique solution if k = -2

(C) The system has unique solution if k = 2

(D) The system has no solution if k = 2

(E) The system has infinite number of solutions if k $\neq -2$.

Which of the following statements are correct?

(A) (B) and (E) only **(B)** (C) and (D) only **(C)** (A) and (E) only **(D)** (A) and (D) only

Q27. MCQ (Single Correct Answer)

Open solution

Let f(x) be a differentiable function defined on [0, 2] such that f'(x) = f'(2 $-$ x) for all x$\in$ (0, 2), f(0) = 1 and f(2) = e^2. Then the value of $\int_0^2 f(x)dx$ is :

(A) 1 + e^2 **(B)** 2(1 + e^2) **(C)** 1 $-$ e^2 **(D)** 2(1 $-$ e^2)

Q28. MCQ (Single Correct Answer)

Open solution

Let f be a twice differentiable function defined on R such that f(0) = 1, f'(0) = 2 and f'(x) $\neq$ 0 for all x $\in$ R. If $\begin{vmatrix} f(x) & f'(x) \\ f'(x) & f''(x) \end{vmatrix}$ = 0, for all x$\in$R, then the value of f(1) lies in the interval :

(A) (0, 3) **(B)** (9, 12) **(C)** (3, 6) **(D)** (6, 9)

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Q29. MCQ (Single Correct Answer)

Let $f : R \to R$ be defined as

$$f(x) = \begin{cases} -55x, & if\, x < -5 \\ 2x^3 - 3x^2 - 120x, & if\, -5 \le x \le 4 \\ 2x^3 - 3x^2 - 36x - 336, & if\, x > 4, \end{cases}$$

Let A = {x ∈ R : f is increasing}. Then A is equal to :

(A) $(-5, \infty)$

(B) $(-\infty, -5) \cup (4, \infty)$

(C) $(-5, -4) \cup (4, \infty)$

(D) $(-\infty, -5) \cup (-4, \infty)$

Open solution

Q30. MCQ (Single Correct Answer)

If P is a point on the parabola y = x^2 + 4 which is closest to the straight line y = 4x − 1, then the co-ordinates of P are :

(A) $(-2, 8)$

(B) (2, 8)

(C) (1, 5)

(D) (3, 13)

Open solution

Q31. MCQ (Single Correct Answer)

The probability that two randomly selected subsets of the set {1, 2, 3, 4, 5} have exactly two elements in their intersection, is :

(A) $\frac{135}{2^9}$

(B) $\frac{65}{2^8}$

(C) $\frac{65}{2^7}$

(D) $\frac{35}{2^7}$

Open solution

Q32. MCQ (Single Correct Answer)

A possible value of $\tan\left(\frac{1}{4}\sin^{-1}\frac{\sqrt{63}}{8}\right)$ is :

(A) $\sqrt{7} - 1$

(B) $\frac{1}{\sqrt{7}}$

(C) $2\sqrt{2} - 1$

(D) $\frac{1}{2\sqrt{2}}$

Open solution

Q33. MCQ (Single Correct Answer)

The area of the region : $R = \{(x, y) : 5x^2 \le y \le 2x^2 + 9\}$ is :

(A) $6\sqrt{3}$ square units

(B) $12\sqrt{3}$ square units

(C) $11\sqrt{3}$ square units

(D) $9\sqrt{3}$ square units

Open solution

Q34. Numerical

Let a point P be such that its distance from the point (5, 0) is thrice the distance of P from the point (−5, 0). If the locus of the point P is a circle of radius r, then $4r^2$ is equal to _____

Open solution

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Q35. Numerical

If the variance of 10 natural numbers 1, 1, 1, ..., 1, k is less than 10, then the maximum possible value of k is _____.

Open solution

Q36. Numerical

Let λ be an integer. If the shortest distance between the lines

$x - \lambda = 2y - 1 = -2z$ and $x = y + 2\lambda = z - \lambda$ is $\frac{\sqrt{7}}{2\sqrt{2}}$, then the value of $|\lambda|$ is _____.

Open solution

Q37. Numerical

If $a + \alpha = 1$, $b + \beta = 2$ and $af(x) + \alpha f\left(\frac{1}{x}\right) = bx + \frac{\beta}{x}, x \neq 0$, then the value of the expression $\frac{f(x)+f\left(\frac{1}{x}\right)}{x+\frac{1}{x}}$ is _____.

Open solution

Q38. Numerical

Let $i = \sqrt{-1}$. If $\frac{\left(-1+i\sqrt{3}\right)^{21}}{(1-i)^{24}} + \frac{\left(1+i\sqrt{3}\right)^{21}}{(1+i)^{24}} = k$, and $n = [|k|]$ be the greatest integral part of $|k|$. Then $\sum_{j=0}^{n+5} (j+5)^2 - \sum_{j=0}^{n+5} (j+5)$ is equal to _____.

Open solution

Q39. Numerical

The number of the real roots of the equation $(x+1)^2 + |x-5| = \frac{27}{4}$ is _____.

Open solution

Q40. Numerical

The students $S_1, S_2, ..., S_{10}$ are to be divided into 3 groups A, B and C such that each group has at least one student and the group C has at most 3 students. Then the total number of possibilities of forming such groups is _____.

Open solution

Q41. Numerical

The sum of first four terms of a geometric progression (G. P.) is $\frac{65}{12}$ and the sum of their respective reciprocals is $\frac{65}{18}$. If the product of first three terms of the G.P. is 1, and the third term is α, then 2α is _____.

Open solution

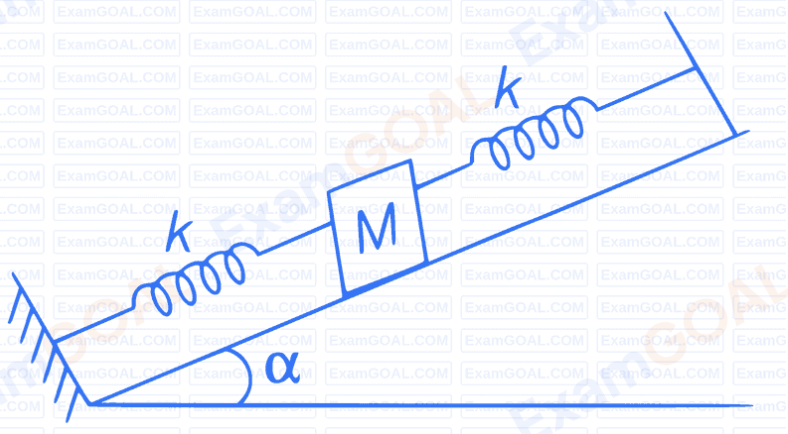
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Physics 27 questions

Q42. MCQ (Single Correct Answer)

In the given figure, a body of mass M is held between two massless springs, on a smooth inclined plane. The free ends of the springs are attached to firm supports. If each spring has spring constant k, the frequency of oscillation of given body is :

Open solution



(A) $\frac{1}{2\pi}\sqrt{\frac{2k}{Mg\sin\alpha}}$

(B) $\frac{1}{2\pi}\sqrt{\frac{k}{Mg\sin\alpha}}$

(C) $\frac{1}{2\pi}\sqrt{\frac{2k}{M}}$

(D) $\frac{1}{2\pi}\sqrt{\frac{k}{2M}}$

Q43. MCQ (Single Correct Answer)

If the source of light used in a Young's double slit experiment is changed from red to violet :

Open solution

(A) the fringes will become brighter.

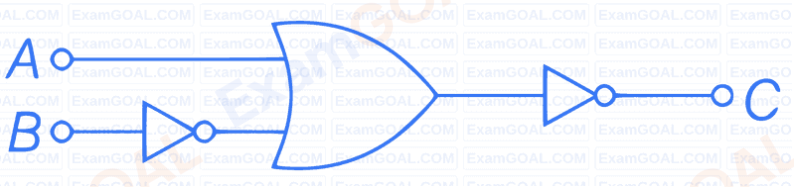
(B) the intensity of minima will increase.

(C) consecutive fringe lines will come closer.

(D) the central bright fringe will become a dark fringe.

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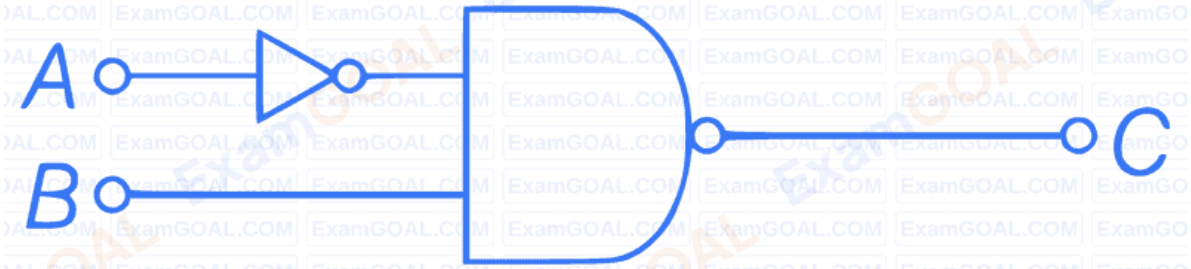
Q44. MCQ (Single Correct Answer)



Open solution

The logic circuit shown above is equivalent to :

(A)



(B)



(C)



(D)

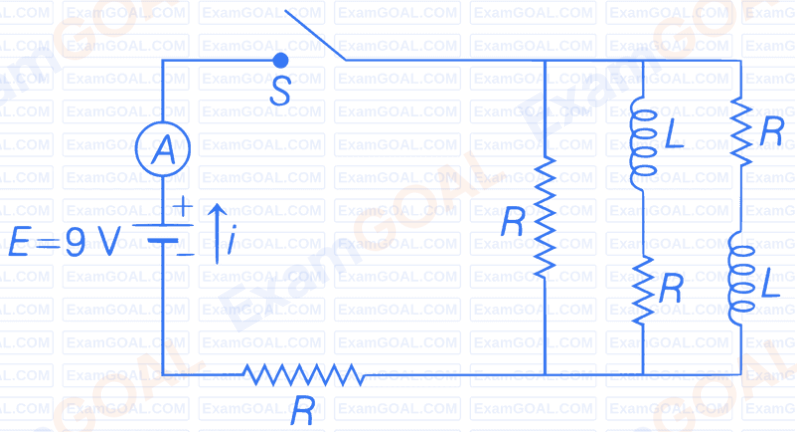


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Q45. MCQ (Single Correct Answer)

Figure shows a circuit that contains four identical resistors with resistance R = 2.0Ω, two identical inductors with inductance L = 2.0 mH and an ideal battery with emf E = 9V. The current 'i' just after the switch 'S' is closed will be :

Open solution



(A) 3.0 A (B) 3.37 A (C) 9 A (D) 2.25 A

Q46. MCQ (Single Correct Answer)

A particle is projected with velocity v_0 along x-axis. A damping force is acting on the particle which is proportional to the square of the distance from the origin i.e. ma = $-\alpha x^2$. The distance at which the particle stops :

Open solution

(A) $\left[\frac{3mv_0^2}{2\alpha}\right]^{\frac{1}{3}}$ (B) $\left(\frac{2v_0}{3\alpha}\right)^{\frac{1}{3}}$

(C) $\left(\frac{3v_0^2}{2\alpha}\right)^{\frac{1}{2}}$ (D) $\left(\frac{2v_0^2}{3\alpha}\right)^{\frac{1}{2}}$

Q47. MCQ (Single Correct Answer)

An X-ray tube is operated at 1.24 million volt. The shortest wavelength of the produced photon will be :

Open solution

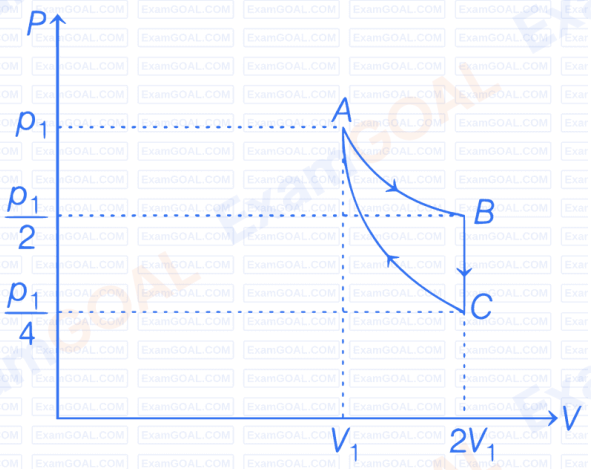
(A) 10^{-2} nm (B) 10^{-1} nm (C) 10^{-3} nm (D) 10^{-4} nm

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Q48. MCQ (Single Correct Answer)

Open solution

If one mole of an ideal gas at (P_1, V_1) is allowed to expand reversibly and isothermally (A to B) its pressure is reduced to one-half of the original pressure (see figure). This is followed by a constant volume cooling till its pressure is reduced to one-fourth of the initial value (B $\rightarrow$ C). Then it is restored to its initial state by a reversible adiabatic compression (C to A). The net work done by the gas is equal to :



(A) $-\frac{RT}{2(\gamma-1)}$

(B) $RT\left(\ln 2-\frac{1}{2(\gamma-1)}\right)$

(C) $RT\ln 2$

(D) 0

Q49. MCQ (Single Correct Answer)

Open solution

A body weights 49N on a spring balance at the north pole. What will be its weight recorded on the same weighing machine, if it is shifted to the equator?

[Use $g=\frac{GM}{R^2}$ = 9.8 ms^{-2} and radius of earth, R = 6400 km.]

(A) 49 N

(B) 49.83 N

(C) 48.83 N

(D) 49.17 N

Q50. MCQ (Single Correct Answer)

Open solution

When a particle executes SHM, the nature of graphical representation of velocity as a function of displacement is :

(A) circular

(B) straight line

(C) parabolic

(D) elliptical

Q51. MCQ (Single Correct Answer)

Open solution

The period of oscillation of a simple pendulum is $T=2\pi\sqrt{\frac{L}{g}}$. Measured value of 'L' is 1.0 m from meter scale having a minimum division of 1 mm and time of one complete oscillation is 1.95 s measured from stopwatch of 0.01 s resolution. The percentage error in the determination of 'g' will be :

(A) 1.30%

(B) 1.33%

(C) 1.13%

(D) 1.03%

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Q52. MCQ (Single Correct Answer)

A soft ferromagnetic material is placed in an external magnetic field. The magnetic domains :

(A) may increase or decrease in size and change its orientation.

(B) increase in size but no change in orientation.

(C) decrease in size and changes orientation.

(D) have no relation with external magnetic field.

Open solution

Q53. MCQ (Single Correct Answer)

Which of the following equations represents a travelling wave?

(A) $y = Ae^{x}\cos(\omega t - \theta)$

(B) $y = Ae^{-x^2}(vt + \theta)$

(C) $y = A \sin(15x - 2t)$

(D) $y = A \sin x \cos \omega t$

Open solution

Q54. MCQ (Single Correct Answer)

Two electrons each are fixed at a distance '2d'. A third charge proton placed at the midpoint is displaced slightly by a distance x (x << d) perpendicular to the line joining the two fixed charges. Proton will execute simple harmonic motion having angular frequency : (m = mass of charged particle)

(A) $\left(\frac{2q^2}{\pi\varepsilon_0 md^3}\right)^{\frac{1}{2}}$

(B) $\left(\frac{q^2}{2\pi\varepsilon_0 md^3}\right)^{\frac{1}{2}}$

(C) $\left(\frac{2\pi\varepsilon_0 md^3}{q^2}\right)^{\frac{1}{2}}$

(D) $\left(\frac{\pi\varepsilon_0 md^3}{2q^2}\right)^{\frac{1}{2}}$

Open solution

Q55. MCQ (Single Correct Answer)

On the basis of kinetic theory of gases, the gas exerts pressure because its molecules :

(A) continuously lose their energy till it reaches wall.

(B) are attracted by the walls of container.

(C) suffer change in momentum when impinge on the walls of container.

(D) continuously stick to the walls of container.

Open solution

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Q56. MCQ (Single Correct Answer)

Open solution

Match List - I with List - II.

	List I		List II
(a)	Source of microwave frequency	(i)	Radioactive decay of nucleus
(b)	Source of infrared frequency	(ii)	Magnetron
(c)	Source of Gamma Rays	(iii)	Inner shell electrons
(d)	Source of X-rays	(iv)	Vibration of atoms and molecules
		(v)	LASER
		(vi)	RC circuit

Choose the correct answer from the options given below :

(A) (a)-(vi), (b)-(v), (c)-(i), (d)-(iv)

(B) (a)-(ii), (b)-(iv), (c)-(i), (d)-(iii)

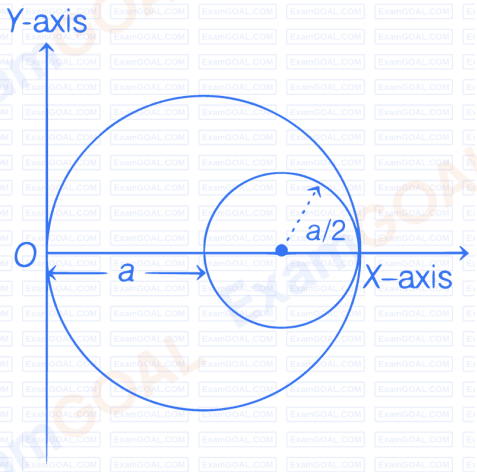
(C) (a)-(ii), (b)-(iv), (c)-(vi), (d)-(iii)

(D) (a)-(vi), (b)-(iv), (c)-(i), (d)-(v)

Q57. MCQ (Single Correct Answer)

Open solution

A circular hole of radius $\left(\frac{a}{2}\right)$ is cut out of a circular disc of radius 'a' as shown in figure. The centroid of the remaining circular portion with respect to point 'O' will be :



(A) $\frac{1}{6}a$

(B) $\frac{2}{3}a$

(C) $\frac{5}{6}a$

(D) $\frac{10}{11}a$

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Q58. MCQ (Single Correct Answer)

Zener breakdown occurs in a p $-$ n junction having p and n both :

(A) heavily doped and have wide depletion layer.

(B) lightly doped and have narrow depletion layer.

(C) heavily doped and have narrow depletion layer.

(D) lightly doped and have wide depletion layer.

Open solution

Q59. MCQ (Single Correct Answer)

The de-Broglie wavelength of a proton and α-particle are equal. The ratio of their velocities is :

(A) 4 : 2 (B) 4 : 3 (C) 4 : 1 (D) 1 : 4

Open solution

Q60. MCQ (Single Correct Answer)

According to Bohr atomic model, in which of the following transitions will the frequency be maximum?

(A) n = 2 to n = 1 (B) n = 3 to n = 2 (C) n = 4 to n = 3 (D) n = 5 to n = 4

Open solution

Q61. Numerical

A uniform metallic wire is elongated by 0.04 m when subjected to a linear force F. The elongation, if its length and diameter is doubled and subjected to the same force will be _____ cm.

Open solution

Q62. Numerical

An electromagnetic wave of frequency 3 GHz enters a dielectric medium of relative electric permittivity 2.25 from vacuum. The wavelength of this wave in that medium will be _____ $\times 10^{-2}$ cm.

Open solution

Q63. Numerical

A cylindrical wire of radius 0.5 mm and conductivity 5×10^7 S/m is subjected to an electric field of 10 mV/m. The expected value of current in the wire will be $x^3\pi$ mA. The value of x is _____.

Open solution

Q64. Numerical

Two solids A and B of mass 1 kg and 2 kg respectively are moving with equal linear momentum. The ratio of their kinetic energies $(K.E.)_A : (K.E.)_B$ will be $\frac{A}{1}$, so the value of A will be _____.

Open solution

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Q65. Numerical

The root mean square speed of molecules of a given mass of a gas at 27°C and 1 atmosphere pressure is 200 ms^{-1}. The root mean square speed of molecules of the gas at 127°C and 2 atmosphere pressure is $\frac{x}{\sqrt{3}}$ ms^{-1}. The value of x will be _____.

Open solution

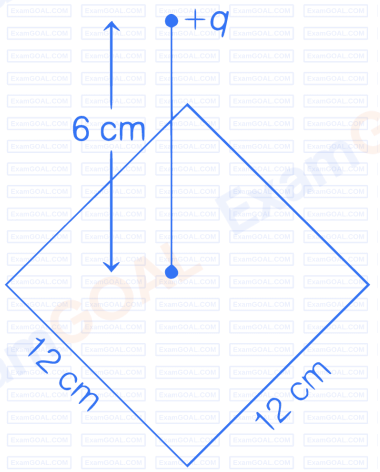
Q66. Numerical

A series L-C-R circuit is designed to resonate at an angular frequency $\omega_0 = 10^5$ rad/s. The circuit draws 16W power from 120V source at resonance. The value of resistance 'R' in the circuit is _____ Ω.

Open solution

Q67. Numerical

A point charge of +12μC is at a distance 6 cm vertically above the centre of a square of side 12 cm as shown in figure. The magnitude of the electric flux through the square will be _____ $\times 10^3$ Nm^2/C.



Open solution

Q68. Numerical

A uniform thin bar of mass 6 kg and length 2.4 meter is bent to make an equilateral hexagon. The moment of inertia about an axis passing through the centre of mass and perpendicular to the plane of hexagon is _____ $\times 10^{-1}$ kg m^2.

Open solution

Out of syllabus 22 questions

These questions are on topics no longer in the current syllabus. They are kept for practice, after everything else.

Q69. MCQ (Single Correct Answer) · **Chemistry**

In polymer Buna-S : 'S' stands for :

(A) Sulphur (B) Styrene (C) Strength (D) Sulphonation

Open solution

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Q70. MCQ (Single Correct Answer) · **Chemistry**

Open solution

Given below are two statements : one is labelled as Assertion A and the other is labelled as Reason R.

Assertion A : Hydrogen is the most abundant element in the Universe, but it is not the most abundant gas in the troposphere.

Reason R : Hydrogen is the lightest element.

In the light of the above statements, choose the correct answer from the options given below :

(A) Both A and R are true but R is NOT the correct explanation of A

(B) Both A and R are true and R is the correct explanation of A

(C) A is true but R is false

(D) A is false but R is true.

Q71. MCQ (Single Correct Answer) · **Chemistry**

Open solution

Given below are two statements :

Statement I : The value of the parameter "Biochemical Oxygen Demand (BOD)" is important for survival of aquatic life.

Statement II : The optimum value of BOD is 6.5 ppm.

In the light of the above statements, choose the most appropriate answer from the options given below :

(A) Statement I is false but Statement II is true

(B) Statement I is true but Statement II is false

(C) Both Statement I and Statement II are true

(D) Both Statement I and Statement II are false

Q72. MCQ (Single Correct Answer) · **Chemistry**

Open solution

Most suitable salt which can be used for efficient clotting of blood will be :

(A) $Mg(HCO_3)_2$ (B) $NaHCO_3$ (C) $FeCl_3$ (D) $FeSO_4$

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Q73. MCQ (Single Correct Answer) · **Chemistry**

Open solution

Math List - I with List - II.

	List I (Metal)		List II (Ores)
(a)	Aluminium	(i)	Siderite
(b)	Iron	(ii)	Calamine
(c)	Copper	(iii)	Kaolinite
(d)	Zinc	(iv)	Malachite

Choose the correct answer from the options given below :

(A) (a)-(iv), (b)-(iii), (c)-(ii), (d)-(i)

(B) (a)-(i), (b)-(ii), (c)-(iii), (d)-(iv)

(C) (a)-(iii), (b)-(i), (c)-(iv), (d)-(ii)

(D) (a)-(ii), (b)-(iv), (c)-(i), (d)-(iii)

Q74. MCQ (Single Correct Answer) · **Chemistry**

Open solution

Match List - I with List - II.

	List I (Salt)		List II (Flame colour wavelength)
(a)	LiCl	(i)	455.5 nm
(b)	NaCl	(ii)	670.8 nm
(c)	RbCl	(iii)	780.0 nm
(d)	CsCl	(iv)	589.2 nm

Choose the correct answer from the options given below :

(A) (a)-(ii), (b)-(i), (c)-(iv), (d)-(iii)

(B) (a)-(iv), (b)-(ii), (c)-(iii), (d)-(i)

(C) (a)-(i), (b)-(iv), (c)-(ii), (d)-(iii)

(D) (a)-(ii), (b)-(iv), (c)-(iii), (d)-(i)

Q75. Numerical · **Chemistry**

Open solution

Among the following allotropic forms of sulphur, the number of allotropic forms, which will show paramagnetism is _____.

(A) α-sulphur

(B) β-sulphur

(C) S_2-form

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Q76. Numerical · **Chemistry**

The volume occupied by 4.75 g of acetylene gas at 50°C and 740 mmHg pressure is _____ L. (Rounded off to the nearest integer)

[Given R = 0.0826 L atm K^{-1} mol^{-1}]

Open solution

Q77. MCQ (Single Correct Answer) · **Mathematics**

If a curve y = f(x) passes through the point (1, 2) and satisfies $x\frac{dy}{dx} + y = bx^4$, then for what value of b, $\int_1^2 f(x)dx = \frac{62}{5}$?

(A) $\frac{31}{5}$ (B) 10 (C) 5 (D) $\frac{62}{5}$

Open solution

Q78. MCQ (Single Correct Answer) · **Mathematics**

For which of the following curves, the line $x + \sqrt{3}y = 2\sqrt{3}$ is the tangent at the point $\left(\frac{3\sqrt{3}}{2}, \frac{1}{2}\right)$?

(A) $2x^2 - 18y^2 = 9$ (B) $y^2 = \frac{1}{6\sqrt{3}}x$ (C) $x^2 + 9y^2 = 9$ (D) $x^2 + y^2 = 7$

Open solution

Q79. MCQ (Single Correct Answer) · **Mathematics**

Let a, b, c be in arithmetic progression. Let the centroid of the triangle with vertices (a, c), (2, b) and (a, b) be $\left(\frac{10}{3}, \frac{7}{3}\right)$. If α, β are the roots of the equation $ax^2 + bx + 1 = 0$, then the value of $\alpha^2 + \beta^2 - \alpha\beta$ is :

(A) $\frac{69}{256}$ (B) $\frac{71}{256}$ (C) $-\frac{71}{256}$ (D) $-\frac{69}{256}$

Open solution

Q80. MCQ (Single Correct Answer) · **Mathematics**

The negation of the statement

$\sim p \wedge (p \vee q)$ is :

(A) $p \vee \sim q$ (B) $\sim p \vee q$ (C) $\sim p \wedge q$ (D) $p \wedge \sim q$

Open solution

Q81. MCQ (Single Correct Answer) · **Mathematics**

For the statements p and q, consider the following compound statements :

(a) $(\sim q \wedge (p \to q)) \to \sim p$

(b) $((p \vee q) \wedge \sim p) \to q$

Then which of the following statements is correct?

(A) (b) is a tautology but not (a).

(B) (a) and (b) both are not tautologies.

(C) (a) and (b) both are tautologies.

(D) (a) is a tautology but not (b).

Open solution

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Q82. MCQ (Single Correct Answer) · **Mathematics**

If $n \geq 2$ is a positive integer, then the sum of the series ${}^{n+1}C_2 + 2\left({}^2C_2 + {}^3C_2 + {}^4C_2 + \ldots + {}^nC_2\right)$ is :

(A) $\frac{n(2n+1)(3n+1)}{6}$

(B) $\frac{n(n+1)(2n+1)}{6}$

(C) $\frac{n(n+1)^2(n+2)}{12}$

(D) $\frac{n(n-1)(2n+1)}{6}$

Open solution

Q83. MCQ (Single Correct Answer) · **Mathematics**

The vector equation of the plane passing through the intersection

of the planes $\vec{r}.\left(\hat{i}+\hat{j}+\hat{k}\right)=1$ and $\vec{r}.\left(\hat{i}-2\hat{j}\right)=-2$, and the point (1, 0, 2) is :

(A) $\vec{r}.\left(\hat{i}+7\hat{j}+3\hat{k}\right)=\frac{7}{3}$

(B) $\vec{r}.\left(\hat{i}+7\hat{j}+3\hat{k}\right)=7$

(C) $\vec{r}.\left(3\hat{i}+7\hat{j}+3\hat{k}\right)=7$

(D) $\vec{r}.\left(\hat{i}-7\hat{j}+3\hat{k}\right)=\frac{7}{3}$

Open solution

Q84. MCQ (Single Correct Answer) · **Mathematics**

The angle of elevation of a jet plane from a point A on the ground is 60°. After a flight of 20 seconds at the speed of 432 km/hour, the angle of elevation changes to 30°. If the jet plane is flying at a constant height, then its height is :

(A) $3600\sqrt{3}$ m

(B) $1200\sqrt{3}$ m

(C) $1800\sqrt{3}$ m

(D) $2400\sqrt{3}$ m

Open solution

Q85. MCQ (Single Correct Answer) · **Mathematics**

If the curve $y = ax^2 + bx + c$, $x \in R$, passes through the point (1, 2) and the tangent line to this curve at origin is y = x, then the possible values of a, b, c are :

(A) a = − 1, b = 1, c = 1

(B) a = 1, b = 1, c = 0

(C) a = $\frac{1}{2}$, b = $\frac{1}{2}$, c = 1

(D) a = 1, b = 0, c = 1

Open solution

Q86. Numerical · **Mathematics**

If the area of the triangle formed by the positive x-axis, the normal and the tangent to the circle $(x-2)^2 + (y-3)^2 = 25$ at the point (5, 7) is A, then 24A is equal to _____.

Open solution

Q87. Numerical · **Mathematics**

For integers n and r, let $\binom{n}{r} = \begin{cases} {}^nC_r, & if\ n \geq r \geq 0 \\ 0, & otherwise \end{cases}$ The maximum value of k for which the sum $\sum_{i=0}^{k}\binom{10}{i}\binom{15}{k-i} + \sum_{i=0}^{k+1}\binom{12}{i}\binom{13}{k+1-i}$ exists, is equal to _____.

Open solution

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Q88. MCQ (Single Correct Answer) · **Physics**

Open solution

Given below are two statements :

Statement I : PN junction diodes can be used to function as transistor, simply by connecting two diodes, back to back, which acts as the base terminal.

Statement II : In the study of transistor, the amplification factor β indicates ratio of the collector current to the base current.

In the light of the above statements, choose the correct answer from the options given below.

(A) Both Statement I and Statement II are false

(B) Statement I is false but Statement II is true

(C) Both Statement I and Statement II are true

(D) Statement I is true but Statement II is false

Q89. Numerical · **Physics**

Open solution

A signal of 0.1 kW is transmitted in a cable. The attenuation of cable is -5 dB per km and cable length is 20 km. The power received at receiver is 10^{-x} W. The value of x is _____.

[Gain in $dB = 10\log_{10}\left(\frac{P_o}{P_i}\right)$]

Q90. Numerical · **Physics**

Open solution

Two cars are approaching each other at an equal speed of 7.2 km/hr. When they see each other, both blow horns having frequency of 676 Hz. The beat frequency heard by each driver will be _____ Hz. [Velocity of sound in air is 340 m/s.]

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Answer Key

Chemistry

1. **D**	2. **D**	3. **D**	4. **D**	5. **C**
6. **D**	7. **C**	8. **B**	9. **B**	10. **C**
11. **D**	12. **A**	13. **C**	14. **A**	15. **8**
16. **141**	17. **243**	18. **855**	19. **1**	20. **81**
21. **3776**	22. **3**			

Mathematics

23. **C**	24. **D**	25. **A**	26. **D**	27. **A**
28. **D**	29. **C**	30. **B**	31. **A**	32. **B**
33. **B**	34. **56**	35. **11**	36. **1**	37. **2**
38. **310**	39. **2**	40. **31650**	41. **3**	

Physics

42. **C**	43. **C**	44. **D**	45. **D**	46. **A**
47. **C**	48. **B**	49. **C**	50. **D**	51. **C**
52. **A**	53. **C**	54. **B**	55. **C**	56. **B**
57. **C**	58. **C**	59. **C**	60. **A**	61. **2**
62. **667**	63. **5**	64. **2**	65. **400**	66. **900**
67. **226**	68. **8**			

Out of syllabus

69. **B**	70. **B**	71. **B**	72. **C**	73. **C**
74. **D**	75. **1**	76. **5**	77. **B**	78. **C**
79. **C**	80. **A**	81. **C**	82. **B**	83. **B**
84. **B**	85. **B**	86. **1225**	87. **12**	88. **B**
89. **8**	90. **8**			

For numerical answers given as a range, any value within the range is correct. Full solutions: scan the QR code next to each question, or visit questions.examside.com.

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